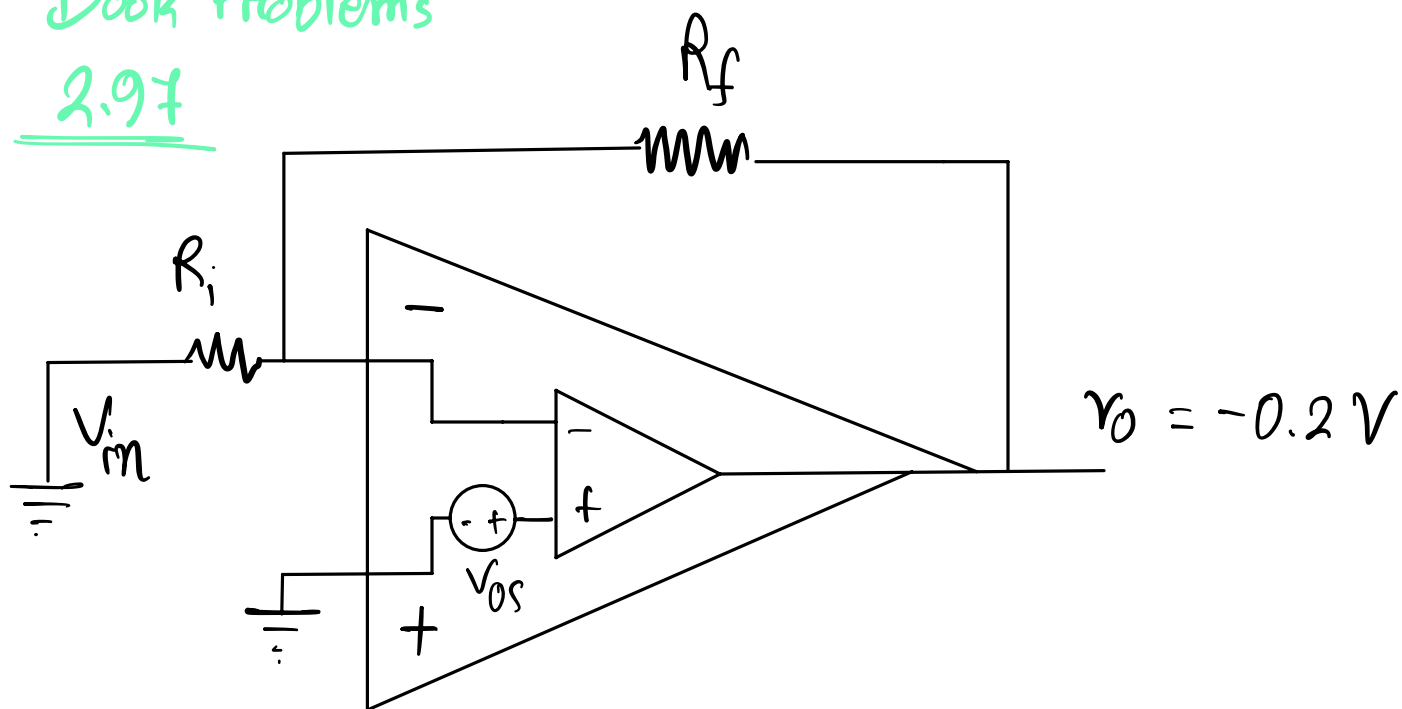


Book Problems

2.97



$$V_{in} = 0$$

$$G = 46 \text{ dB}$$

$$= 10^{\frac{46}{20}}$$

$$= 199.53 \text{ V/V}$$

In an inverting amp, $|G| = \frac{R_f}{R_i} = 199.53 \text{ V/V}$

In case of there being an input offset voltage,

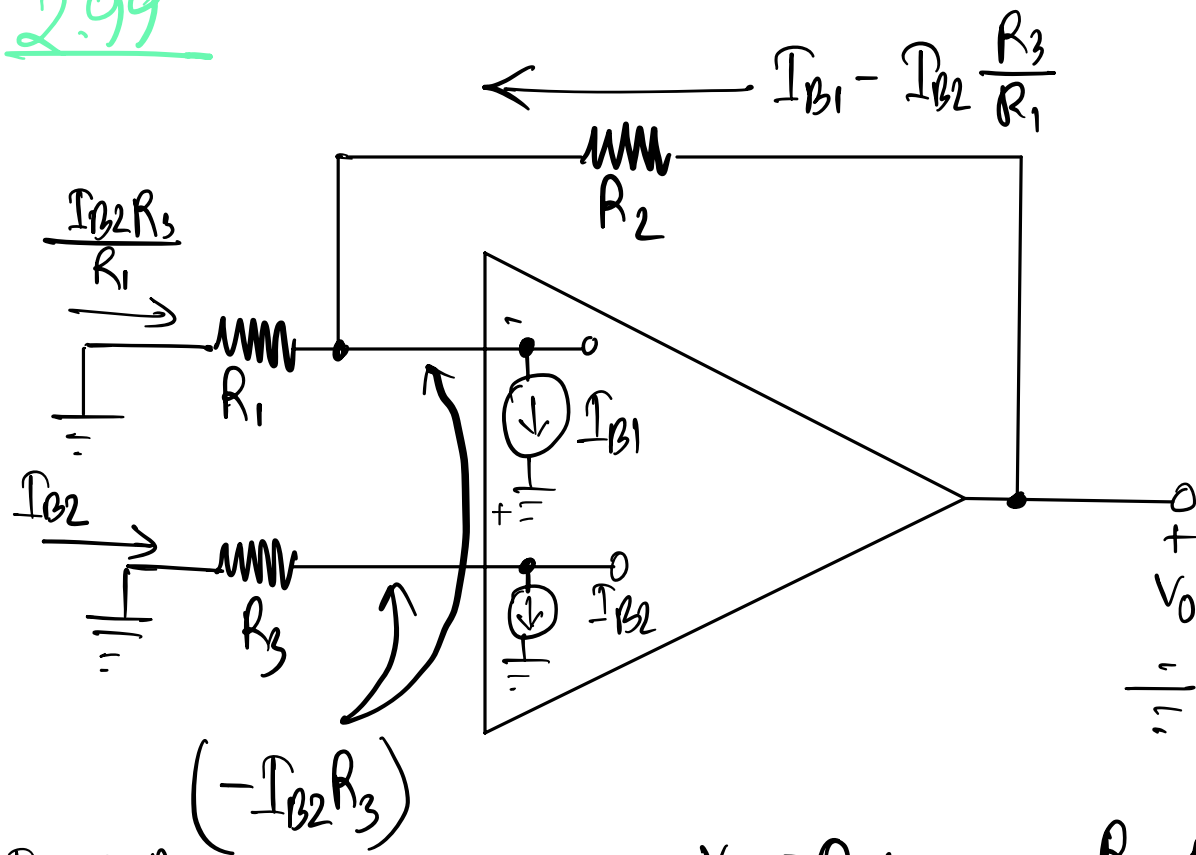
$$V_o = \left(-\frac{R_f}{R_i}\right)V_i + \left(1 + \frac{R_f}{R_i}\right)V_{os}$$

$$\Rightarrow -0.2 = 0 + 200.53 V_{os}$$

$$\Rightarrow V_{os} = 0.997 \text{ mV}$$

$$\Rightarrow V_{os} \approx 1 \text{ mV}$$

2.99



$$I_B = 2 \mu A$$

$$I_{OS} = 0.2 \mu A = |I_{B1} - I_{B2}|$$

So, possible pairs $(I_{B1}, I_{B2}) =$

$$I_{B1} = 2.1 \mu A, I_{B2} = 1.9 \mu A$$

$$I_{B1} = 1.9 \mu A, I_{B2} = 2.1 \mu A$$

KCL at inverting node,

$$\frac{V_n}{R_1} + \frac{V_n - V_0}{R_2} + I_{B1} = 0$$

$$\Rightarrow \frac{V_0}{R_2} = I_{B1} + \left(\frac{1}{R_1} + \frac{1}{R_2} \right) V_n = I_{B1} + \left(\frac{1}{R_1} + \frac{1}{R_2} \right) (-I_{B2} R_3)$$

$$\Rightarrow V_0 = I_{B1} R_2 - \left(1 + \frac{R_2}{R_1} \right) I_{B2} R_3$$

$$\Rightarrow V_0 = 100K I_{B1} - 500K I_{B2}$$

$$R_1 = 11.1K \Omega$$

$$R_2 = 100K \Omega$$

$$R_3 = 5K \Omega$$

$$\left\{ \begin{aligned} 1 + \frac{R_2}{R_1} &= 1 + \frac{100K}{11.1K} \\ &= 10.009 \\ &\approx 10 \end{aligned} \right.$$

Case 1: $I_{B1} = 2.1 \mu A$, $I_{B2} = 1.9 \mu A$

$$V_o = 0.115 V$$

Case 2: $I_{B1} = 1.9 \mu A$, $I_{B2} = 2.1 \mu A$

$$V_o = 0.085 V$$

Now,

$$R_n = R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2} = 9.99 k\Omega \approx 10 k\Omega$$

so, R_p has to be $10 k\Omega$

we have $R_3 = 5 k\Omega$

$$\text{so, } R_p = 5 k\Omega + R$$

$\Rightarrow R = 5 k\Omega$ in series

After adding $5 k\Omega$ in series at the non-inverting end with $5 k\Omega$,

$$V_o = R_2 (I_{B1} - I_{B2})$$

$$|V_o| = 100 k\Omega \cdot 0.2 \mu A = 0.02 V$$

so, range becomes $-0.02 V \leq V_o \leq 0.02 V$

We know,

$$R_1 \parallel R_2 = 10k$$

If the designer wants $15k$ source resistor,
we need to add R in parallel to reduce
it,

$$\frac{1}{10k} = \frac{1}{15k} + \frac{1}{R}$$

$$\Rightarrow R = 30k\Omega$$

So, we need to add $30k\Omega$ in parallel

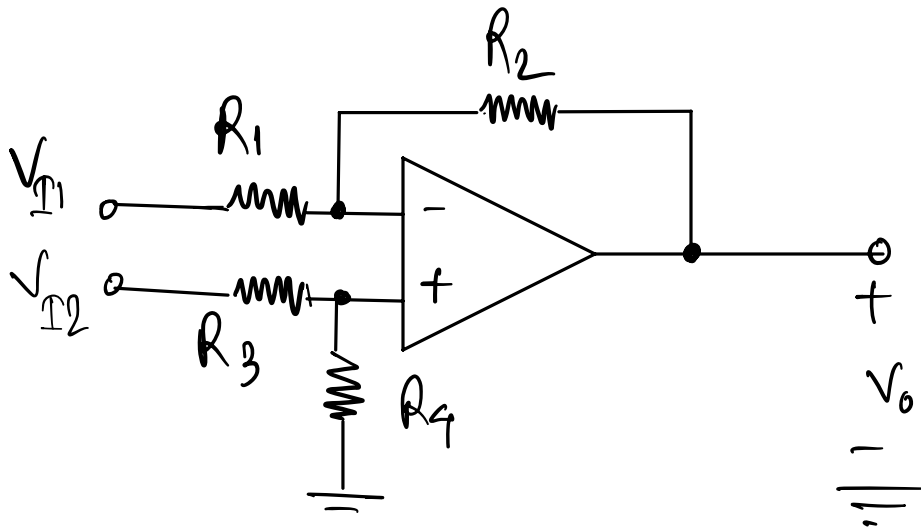
So, with $5k\Omega$ source, the voltage range,

$$0.085V \leq V_o \leq 0.115V$$

We need to add a $5k\Omega$ resistor with the $5k\Omega$
to compensate for the bias-current

And to use with a $15k\Omega$ source, we need to put
a $30k\Omega$ resistor in parallel.

2.101



In the differential amplifier,

$$V_{I1} = V_{I2} = 0 \text{ (Ideal source)}$$

$$R_1 = R_3 = 100 \text{ k}\Omega$$

$$R_2 = R_4 = 1 \text{ M}\Omega$$

$$V_{OS} = 5 \text{ mV}$$

$$I_B = 100 \text{ nA}$$

$$I_{OS} = 20 \text{ nA}$$

we know,

$$V_n = V_p + V_{OS}$$

KCL at non-inverting node,

$$\frac{V_p}{R_3} + \frac{V_p}{R_4} + I_{B2} = 0$$

$$\Rightarrow V_p = - \frac{I_{B2}}{(1/R_3 + 1/R_4)}$$

KCL at inverting node,

$$\frac{V_n}{R_1} + \frac{V_n - V_o}{R_2} + I_{B1} = 0$$

$$\Rightarrow V_o = \left(1 + \frac{R_2}{R_1}\right) V_n + R_2 I_{B1}$$

$$\Rightarrow V_o = \left(1 + \frac{R_2}{R_1}\right) (V_p + V_{oc}) + R_2 I_{B1}$$

$$\Rightarrow V_o = \left(1 + \frac{R_2}{R_1}\right) V_{os} + R_2 I_{B1} - \left(1 + \frac{R_2}{R_1}\right) \left(\frac{I_{B2}}{\frac{1}{R_3} + \frac{1}{R_4}} \right)$$

$$\Rightarrow V_o = R_2 I_{B1} - 11 \left(\frac{I_{B2}}{\frac{1}{R_3} + \frac{1}{R_4}} \right) + 11 V_{os}$$

$$\Rightarrow V_o = R_2 I_{B1} - 1 \times 10^6 I_{B2} + 11 (5 \text{ mV})$$

$$\Rightarrow V_o = R_2 I_{B1} - R_2 I_{B2} + 55 \text{ mV}$$

$$\Rightarrow V_o = R_2 (I_{B1} - I_{B2}) + 55 \text{ mV}$$

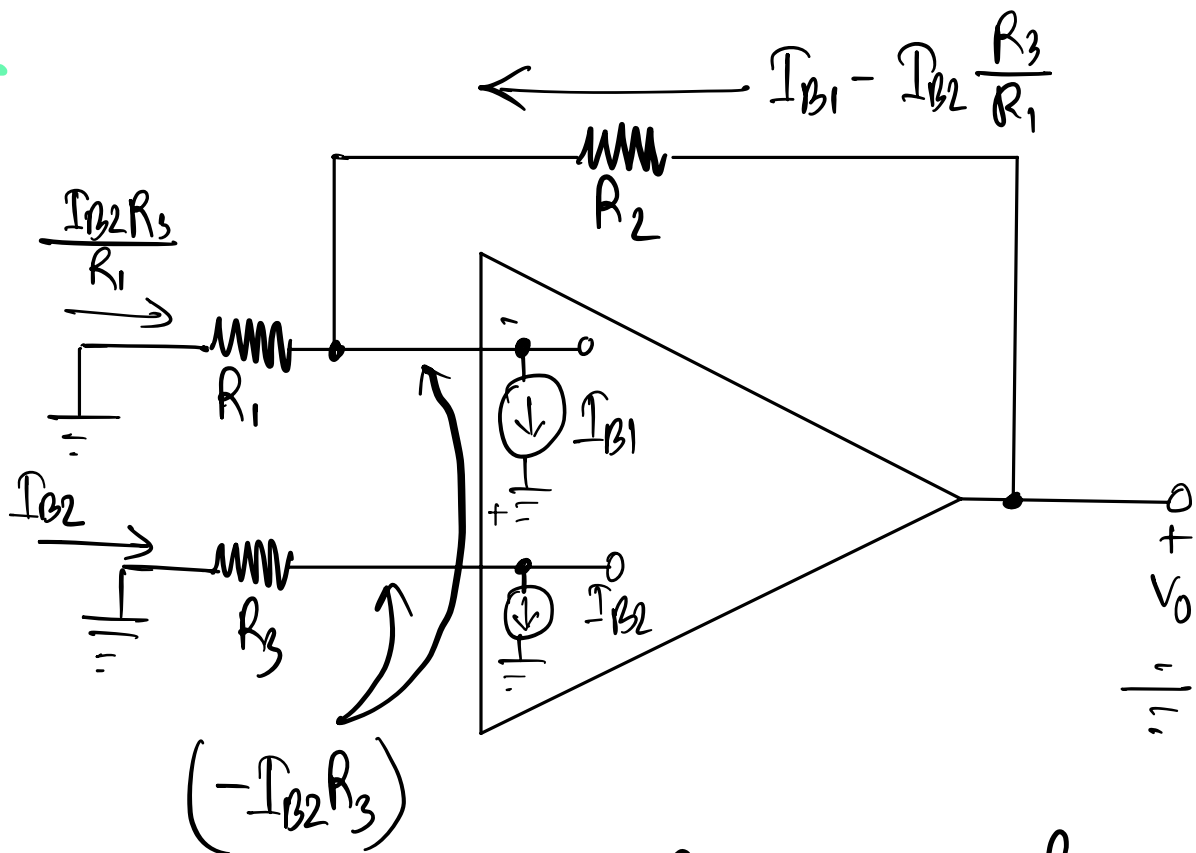
At worst case,

$$I_{B1} - I_{B2} = I_{os} = 20 \text{ nA}$$

$$\begin{aligned} \text{So, } V_o &= (R_2 \times 20 \text{ nA}) + 55 \text{ mV} \\ &= 75 \text{ mV} \end{aligned}$$

So, largest dc output offset 75 mV

2.10D



$$R_1 = 10 \text{ k}\Omega \quad \frac{R_2}{R_1} = 100 \Omega$$

$$R_2 = 1 \text{ m}\Omega$$

For bias current compensation,

$$R_3 = R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2} = 9.90 \text{ k}\Omega$$

With input grounded,

$$V_i = 0 \text{ V}$$

$$V_p = -I_{B2}R_3$$

$$V_0 = 0.30 \text{ V}$$

$$V_p = V_n$$

KCL at the inverting node,

$$\frac{V_n}{R_1} + \frac{V_n - V_0}{R_2} + I_{B1} = 0$$

$$\Rightarrow V_0 = R_2 \left(I_{B1} + \left(\frac{1}{R_1} + \frac{1}{R_2} \right) (-I_{B2} R_3) \right)$$

$$\Rightarrow V_0 = I_{B1} R_2 - \left(1 + \frac{R_2}{R_1} \right) I_{B2} R_3$$

For bias compensation,

$$R_3 = R_1 \parallel R_2$$

$$\text{and } \left(1 + \frac{R_2}{R_1} \right) R_3 = 0.99 \text{ M}\Omega \\ \approx R_2$$

$$\text{So, } V_0 = I_{B1} R_2 - I_{B2} R_2$$

$$\Rightarrow V_0 = R_2 (I_{B1} - I_{B2})$$

$$\Rightarrow V_0 = R_2 I_{OS}$$

$$\Rightarrow I_{OS} = \frac{V_0}{R_2} = \frac{0.30 \text{ V}}{1 \times 10^6 \Omega}$$

$$\Rightarrow I_{OS} = 300 \text{ nA}$$

Now,

$$\text{if } I_{OS} = \frac{1}{10} I_B$$

$$\Rightarrow |I_{B1} - I_{B2}| = \frac{1}{10} I_B$$

we know,

$$I_B = \frac{I_{B1} + I_{B2}}{2}$$

$$\text{Let, } I_{B1} = I_B + \Delta, I_{B2} = I_B - \Delta$$

$$\text{If } V_0 = 0,$$

$$0 = I_{B1} R_2 - \left(1 + \frac{R_2}{R_1} \right) I_{B2} R_3$$

$$\Rightarrow R_3 = \frac{I_{B1} R_2}{\left(1 + \frac{R_2}{R_1} \right) I_{B2}}$$

$$= \frac{R_2}{1 + R_2/R_1} \cdot \frac{I_{B1}}{I_{B2}}$$

$$\text{So, } I_{os} = (I_B + \Delta - I_B + \Delta)$$

$$\Rightarrow 2\Delta = \frac{1}{10} I_B$$

$$\Rightarrow \Delta = \frac{1}{20} I_B$$

$$\text{That means, } \frac{I_{B1}}{I_{B2}} = \frac{I_B + \frac{1}{20} I_B}{I_B - \frac{1}{20} I_B} = \frac{\frac{21}{20}}{\frac{19}{20}} = \frac{21}{19}$$

$$\text{So, } R_3 = \frac{R_L}{1 + R_2/R_1} \cdot \frac{21}{19}$$

$$= 10.9 \text{ k}\Omega$$

Thus,

Assuming ideal voltage source at the input,

R_3 should be $9.90 \text{ k}\Omega$

The input offset current when input is grounded,

$$I_{os} = 300 \text{ nA}$$

$R_3 = 10.9 \text{ k}\Omega$ will result in 0 offset voltage

In a single pole, internally compensated op-amp,

$$A = \frac{A_0}{\sqrt{1 + (f/f_p)^2}}$$

Draining by 20dB,

$$A = \frac{A_0}{10} = \frac{A_0}{\sqrt{1 + (f/f_p)^2}}$$

$$\Rightarrow 10 = \sqrt{1 + (f/f_p)^2}$$

$$\Rightarrow f_p = \frac{f}{\sqrt{99}}$$

and, unity-gain frequency,

$$f_T = A_0 f_p = A_0 \frac{f}{\sqrt{99}}$$

$$a) A_0 = 2 \times 10^5 \text{ V/V}, \quad f = 5 \times 10^2 \text{ Hz}$$

$$\text{so, } f_{3dB} = 50.25 \text{ Hz}$$

$$f_T = 10.05 \text{ MHz}$$

$$b) A_0 = 20 \times 10^3 \text{ V/V}, \quad f = 1 \text{ kHz}$$

$$f_{3dB} = 100.5 \text{ Hz}$$

$$f_T = 2.01 \text{ MHz}$$

$$c) A_0 = 1800 \text{ V/V}, \quad f = 0.1 \text{ MHz}$$

$$f_{3dB} = 10.05 \text{ kHz}$$

$$f_T = 18.09 \text{ MHz}$$

$$d) A_0 = 60 \text{ dB}, \quad f = 10 \text{ MHz}$$

$$= 10^{60/20} \text{ V/V}$$

$$= 1000 \text{ V/V}$$

$$f_{3dB} = 1.005 \text{ MHz}$$

$$f_T = 1.005 \text{ GHz}$$

$$e) A_0 = 200 \text{ V/mV}, \quad f = 25 \text{ kHz}$$

$$= 200 \times 10^3 \text{ V/V}$$

$$f_{3dB} = 2512.6 \text{ Hz}$$

$$f_T = 502.51 \text{ MHz}$$

Additional Problems

Problem 1

$$V_S = 0 \text{ V} \quad R_f = 20 \text{ k}\Omega \quad R_s = 2 \text{ k}\Omega$$

$$V_o = -80 \text{ mV}$$

$$V_{os} = ?$$

We know,

$$V_o = -\frac{R_f}{R_s} V_S + \left(1 + \frac{R_f}{R_s}\right) V_{os}$$

$$\Rightarrow V_o = \left(1 + \frac{R_f}{R_s}\right) V_{os}$$

$$\Rightarrow V_{os} = \frac{V_o}{\left(1 + R_f/R_s\right)}$$

$$\Rightarrow V_{os} = -7.3 \text{ mV}$$

Problem 2

Given,

$$R_f = 10 \text{ k}\Omega$$

$$R_s = 1 \text{ k}\Omega$$

$$C = 1 \mu\text{F}$$

$$V_{CC} = +8 \text{ V}$$

$$V_{EE} = -8 \text{ V}$$

$$i_p = -20 \text{ nA}$$

$$i_n = -18 \text{ nA}$$

a) KCL at non-inverting node,

$$C \cdot \frac{dv_p}{dt} + i_p = 0$$

$$\Rightarrow \frac{dv_p}{dt} = - \frac{i_p}{C}$$

$$\Rightarrow \int_0^t \frac{dv_p}{dt} dt = \int_0^t - \frac{i_p}{C} dt$$

$$\Rightarrow v_p(t) - v_p(0) = - \frac{i_p}{C} t$$

$$\Rightarrow v_p(t) = v_p(0) - \frac{i_p}{C} t$$

assuming $v_p(0) = 0$,

$$v_p(t) = - \frac{(-20 \text{ nA})}{1 \mu\text{F}} t$$

$$\Rightarrow v_p(t) = 0.02 t \quad \checkmark$$

we know, $v_p = v_n$

KCL at inverting node,

$$\frac{v_n}{R_s} + \frac{v_n - v_o}{R_f} + i_n = 0$$

$$\Rightarrow v_o = \left(1 + \frac{R_f}{R_s}\right) v_n + R_f i_n$$

$$\Rightarrow v_o(t) = \left(1 + \frac{R_f}{R_s}\right) v_p(t) + R_f i_n$$

$$\Rightarrow v_o(t) = \left(1 + \frac{10k}{1k}\right) v_p(t) + (10k)(-18nA)$$

$$\Rightarrow v_o(t) = 11 v_p(t) - 180\mu V$$

$$\text{so, } v_p(t) = -\frac{i_p}{C} t \quad \& \quad v_o(t) = 11 v_p(t) - 180\mu V$$

$$b) v_o(10) = 11 v_p(10) - 180\mu V$$

$$= 11 \left(-\frac{i_p}{C} 10\right) - 180\mu V$$

$$= 2.20 V$$

$$\text{so, } v_o(10) = 2.20 V$$

$$c) v_p(t) = -\frac{i_p}{C} t = 0.02t \quad v > 0$$

so, it ramps upwards, and hits the positive rail first

$$\text{Upper limit is, } 8V - 1.5V = 6.5V$$

$$\text{So, } v_o = 6.5V = 11 \cdot (0.22t) - 180\mu V$$

$$\Rightarrow t = 29.5s$$

It will take 29.5s to saturate.

$$d) \text{ Input bias current, } I_b = \frac{i_p + i_n}{2}$$

$$= \frac{-20 - 18}{2}$$

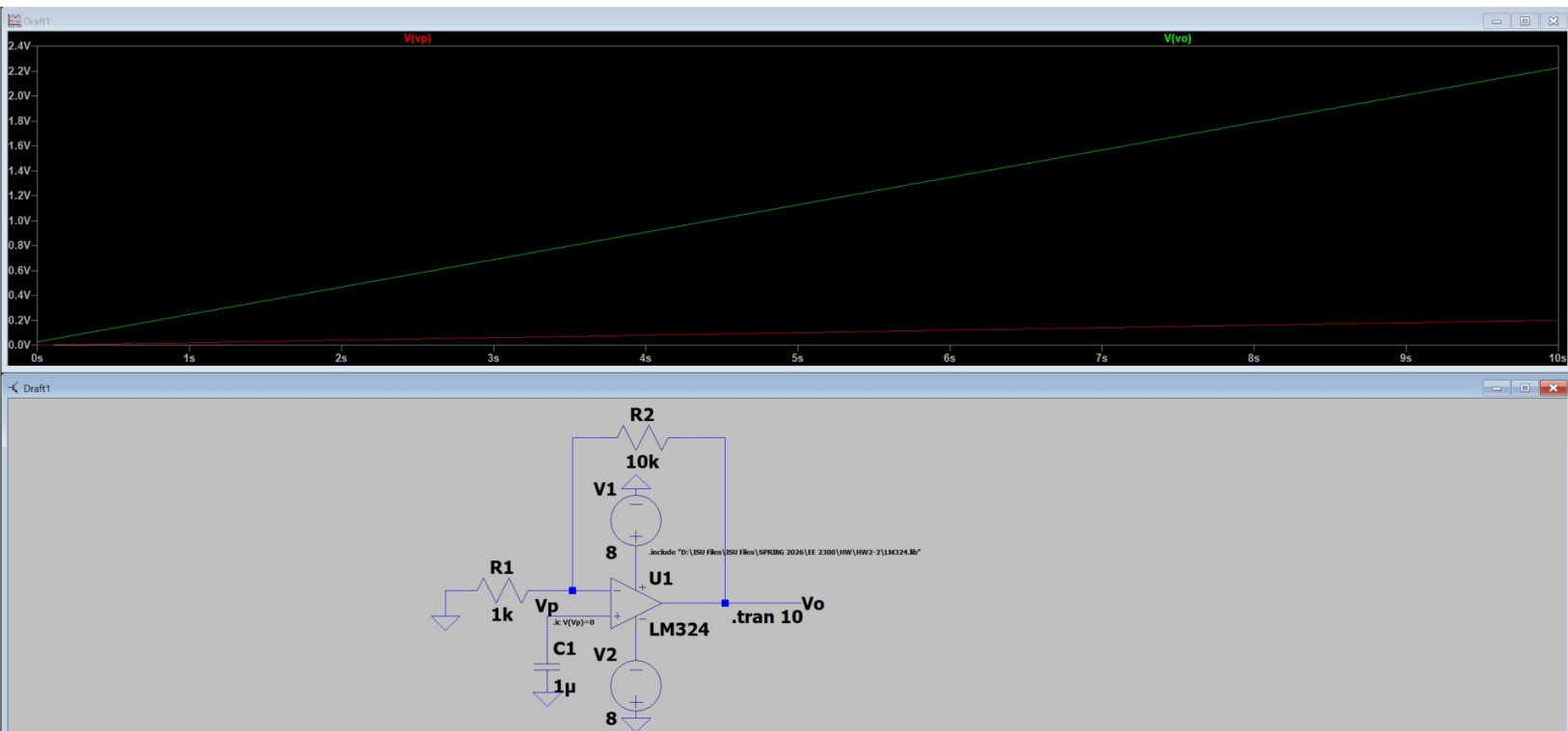
$$= -19 \text{ nA} //$$

$$e) \text{ Input offset current, } I_{os} = |i_p - i_n|$$

$$= |-20 - (-18)|$$

$$= 2 \text{ nA} //$$

f)



Problem 3

Given,

$$R_f = 49 \text{ k}\Omega$$

$$R_s = 1 \text{ k}\Omega$$

so, gain of non-inverting amp,

$$A_o = (1 + R_f/R_s) = 50$$

input sine waves,

$$\text{amplitude } V_{i,\text{peak}} = 0.1 \text{ V}$$

$$\text{frequency} = 50 \text{ kHz}$$

so, output voltage peak amplitude,

$$V_{o,\text{peak}} = A_o V_{i,\text{peak}}$$

$$= 50 \times 0.1 \text{ V}$$

$$= 5 \text{ V}$$

a) We know,

in a closed loop configuration,

$$f_{cl} = \frac{\text{GBW}}{A_o}$$

and f_{cl} has to be at least equal to the

input frequency

$$\text{so, } f_{c1} \geq 50 \text{ kHz}$$

$$\Rightarrow \frac{\text{GBW}}{A_0} \geq 50 \text{ kHz}$$

$$\Rightarrow \text{GBW} \geq A_0 \times 50 \text{ kHz}$$

$$\Rightarrow \text{GBW} \geq 2.5 \times 10^6 \text{ Hz}$$

$$\Rightarrow \text{GBW} \geq 2.5 \text{ MHz}$$

Thus, OA1 is not an option ($\text{GBW} = 1 \text{ MHz}$)

That leaves us with OA2 and OA3

We need maximum power on R_L

$$P_L = \frac{V_L i_L}{2}$$

$$\text{in this case, } P_L = \frac{V_{\text{peak}} i_{o,\text{max}}}{2} \quad (\text{from b})$$

$$\text{so, } R_L \propto i_{o,\text{max}}$$

$$\text{OA2 } i_{o,\text{max}} > \text{OA3 } i_{o,\text{max}}$$

so, I have to choose OA2

b) With OA2,

peak output current, $I_{peak} \leq I_{o,max}$

$$\Rightarrow \frac{V_{peak}}{R_L} \leq I_{o,max}$$

so, at maximum power,

$$R_L = \frac{V_{peak}}{I_{o,max}}$$

Again,

$$P_{max} = \frac{V_{rms}^2}{R_L} = \frac{V_{peak}^2}{2R_L}$$

$$= \frac{V_{peak}^2}{2 \left(\frac{V_{peak}}{I_{o,max}} \right)}$$

$$= \frac{V_{peak} I_{o,max}}{2}$$

for OA2 $i_{o,max} = 75 \text{ mA}$

$$\text{so, } P_{max} = \frac{8 \text{ V} \times 75 \text{ mA}}{2} = 0.1875 \text{ W}$$

c) When choosing the op amp,

I figured that our closed loop frequency has to be at least equal to the input frequency

$$\text{so, } f_c \geq 50 \text{ kHz}$$

from this, I found that the minimum GBW I need is 2.5 MHz

Only OA2 and OA3 satisfy this requirement

Then, we also need maximum power at R_L

For which, I derived $P_{\max} \propto I_{o,\max}$ in (b)

and OA2 has bigger $i_{o,\max}$ than OA3

so, I chose OA2